Stabilization of a Cart-Pendulum System Using Energy-Based Swing-Up Control Under Restricted Track Length

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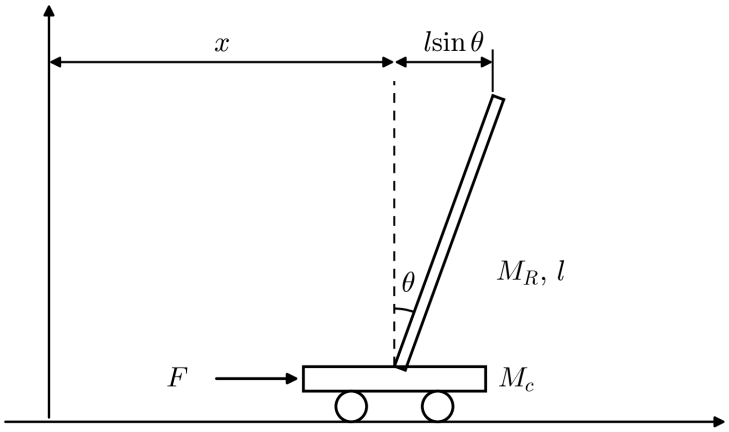


Figure 1: Free body diagram of the cart-pendulum system.

Nomenclature

Symbol	Description
M_R	Mass of rod
M_c	Mass of cart
l	Length of rod
I	Moment of inertia of pendulum rod
g	Acceleration due to gravity
T	Total kinetic energy
U	Total potential energy
L	Length of track
E_{up}	Desired upright energy

1. Modelling Using Lagrangian Mechanics

The Lagrangian for the system can be defined by

$$\mathcal{L} = T - U.$$

The total kinetic energy of the system is

$$T = \tfrac{1}{2}M_R\dot{x}_p^2 + \tfrac{1}{2}M_R\dot{y}_p^2 + \tfrac{1}{2}M_c\dot{x}^2 + \tfrac{1}{2}I\dot{\theta}^2$$

$$T = \tfrac{1}{2}M_R\left(\frac{d}{dt}\Big(x + l\sin\theta\Big)\right)^2 + \tfrac{1}{2}M_R\left(\frac{d}{dt}\Big(l\cos\theta\Big)\right)^2 + \tfrac{1}{2}M_c\dot{x}^2 + \tfrac{1}{2}I\dot{\theta}^2$$

$$T = \tfrac{1}{2}M_R\Big(\dot{x} + l\cos\theta\,\dot{\theta}\Big)^2 + \tfrac{1}{2}M_R\Big(-l\sin\theta\,\dot{\theta}\Big)^2 + \tfrac{1}{2}M_c\dot{x}^2 + \tfrac{1}{2}I\dot{\theta}^2$$

$$T = \tfrac{1}{2}M_R\dot{x}^2 + \tfrac{1}{2}M_Rl^2\cos^2\theta\,\dot{\theta}^2 + M_Rl\dot{x}\cos\theta\,\dot{\theta} + \tfrac{1}{2}M_Rl^2\sin^2\theta\,\dot{\theta}^2 + \tfrac{1}{2}M_c\dot{x}^2 + \tfrac{1}{2}I\dot{\theta}^2$$

$$T = \tfrac{1}{2}M_Rl^2\dot{\theta}^2 + \tfrac{1}{2}M_R\dot{x}^2 + \tfrac{1}{2}M_c\dot{x}^2 + M_Rl\dot{x}\cos\theta\,\dot{\theta} + \tfrac{1}{2}I\dot{\theta}^2 \tag{1}$$

The potential energy is

$$U = M_R\,g\,l\cos\theta. \tag{2}$$

Using (1) and (2),

$$\mathcal{L} = \tfrac{1}{2}M_Rl^2\dot{\theta}^2 + \tfrac{1}{2}M_R\dot{x}^2 + \tfrac{1}{2}M_c\dot{x}^2 + M_Rl\dot{x}\cos\theta\,\dot{\theta} + \tfrac{1}{2}I\dot{\theta}^2 - M_R\,g\,l\cos\theta. \tag{3}$$

The differential equations are obtained as

$$\frac{d}{dt}\left(\frac{\partial\mathcal{L}}{\partial\dot{x}}\right) - \frac{\partial\mathcal{L}}{\partial x} = F,$$

$$\frac{d}{dt}\left(\frac{\partial\mathcal{L}}{\partial\dot{\theta}}\right) - \frac{\partial\mathcal{L}}{\partial\theta} = 0.$$

For the x coordinate,

$$\frac{d}{dt}\Big((M_c + M_R)\dot{x} + M_Rl\cos\theta\,\dot{\theta}\Big) - \underbrace{\frac{\partial\mathcal{L}}{\partial x}}_{0} = F,$$

$$(M_c + M_R)\ddot{x} + M_Rl\cos\theta\,\ddot{\theta} - M_Rl\sin\theta\,\dot{\theta}^2 = F.$$

For the θ coordinate,

$$\frac{d}{dt}\Big((M_Rl^2 + I)\dot{\theta} + M_Rl\dot{x}\cos\theta\Big) - \frac{\partial}{\partial\theta}\Big(M_Rl\dot{x}\cos\theta\,\dot{\theta} - M_R\,g\,l\cos\theta\Big) = 0,$$

$$(M_Rl^2 + I)\ddot{\theta} + M_Rl\ddot{x}\cos\theta - M_Rl\dot{x}\sin\theta\,\dot{\theta} + M_Rl\dot{x}\sin\theta\,\dot{\theta} - M_R\,g\,l\sin\theta = 0,$$

$$\therefore\ (M_Rl^2 + I)\ddot{\theta} + M_Rl\ddot{x}\cos\theta - M_R\,g\,l\sin\theta = 0.$$

The final equations of motion are therefore

$$F = (M_c + M_R)\ddot{x} + M_Rl\cos\theta\,\ddot{\theta} - M_Rl\sin\theta\,\dot{\theta}^2,$$

$$0 = (M_Rl^2 + I)\ddot{\theta} + M_Rl\ddot{x}\cos\theta - M_R\,g\,l\sin\theta.$$

2. Non-Linear State-Space Representation

Define the state vector

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} x \\ \theta \\ \dot{x} \\ \dot{\theta} \end{bmatrix}.$$

The non-linear state-space model can then be obtained as

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} x_3 \\ x_4 \\ \dfrac{\dfrac{F}{M_c + M_R} - \dfrac{3mg \sin x_2 \cos x_2}{4(M_c + M_R)} + \dfrac{M_R l x_4^2 \sin x_2}{M_c + M_R}}{1 - \dfrac{3M_R \cos^2 x_2}{4(M_c + M_R)}} \\ \dfrac{\dfrac{3g \sin x_2}{4l} - \dfrac{3F \cos x_2}{4l(M_c + M_R)} - \dfrac{3M_R x_4^2 \sin x_2 \cos x_2}{4(M_c + M_R)}}{1 - \dfrac{3M_R \cos^2 x_2}{4(M_c + M_R)}} \end{bmatrix}$$

3. Energy-Based Swing-Up Law

Initially, the pendulum is at $\theta = 0$, which is the downward position. The goal is to swing it up by pumping energy until it reaches $\theta = \pi$, which is the upright position.

The energy of the pendulum can be expressed as the sum of its rotational kinetic energy and potential energy. The goal can be achieved by increasing the rotational kinetic energy until the pendulum reaches the upright position.

3.1 Energy-Based Control Law

Let E_{rp} be the energy of the pendulum. This can be expressed as

$$E_{rp} = \tfrac{1}{2} I \dot{\theta}^2 + M_R \, g \, l \cos\theta,$$

$$\frac{dE_{rp}}{dt} = -M_R \ddot{x} \, l \, \dot{\theta} \cos\theta > 0. \tag{4}$$

It can be inferred that the acceleration $\ddot{x}$ and $\dot{\theta}\cos\theta$ must be of opposite sign in order to increase the energy.

Let E_{up} be the desired energy at the upright position. A control strategy can be obtained by the Lyapunov method. A law for the acceleration can be formulated as

$$\ddot{x} = K_{su}\,(E_{rp} - E_{up})\ \mathrm{sgn}(\dot{\theta}\cos\theta). \tag{5}$$

The Lyapunov function $V(t)$ can be formulated as

$$V = \frac{(E_{rp} - E_{up})^2}{2} = \frac{\left(\frac{1}{2}I\dot{\theta}^2 + M_R\,g\,l\cos\theta - 2M_R\,g\,l\right)^2}{2},$$

$$\dot{V} = \frac{2\,(E_{rp} - E_{up})\left(\dot{E}_{rp} - \overset{0}{\dot{E}_{up}}\right)}{2} = (E_{rp} - E_{up})\,\dot{E}_{rp},$$

$$\dot{V} = (E_{rp} - E_{up})\left(-M_R\ddot{x}\,l\,\dot{\theta}\cos\theta\right).$$

Substituting (5),

$$\dot{V} = -K_{su}M_R\,l\,(E_{rp} - E_{up})^2\,|\dot{\theta}\cos\theta|.$$

Checking the Lyapunov conditions,

$$\begin{aligned} &V = 0 \iff E_{rp} = E_{up},\\ &V > 0 \quad \text{whenever } E_{rp} \neq E_{up},\\ &\dot{V} \leq 0 \quad \forall\, t \geq 0. \end{aligned}$$

The Lyapunov function decreases as long as $\dot{\theta} \neq 0$ and $\cos\theta \neq 0$; hence the energy is driven towards the desired value of E_{up}.

3.2 Drawbacks of the Pure Energy Law

From the previous section, the change in energy was defined as

$$\frac{dE_{rp}}{dt} = -M_R\ddot{x}\,l\,\dot{\theta}\cos\theta,$$

which, on substituting the control law, gives

$$\dot{E}_{rp} = -K_{su}M_R\,l\,|\dot{\theta}\cos\theta|\,(E_{rp} - E_{up}).$$

Near the targeted energy, the $|\dot{\theta}\cos\theta|$ term becomes nearly constant. Hence,

$$\dot{E}_{rp} = -C\,(E_{rp} - E_{up}), \qquad \text{where } C > 0 \text{ is some constant.}$$

Let $z = E_{rp} - E_{up}$. When the targeted energy is achieved, $0 < |z| \ll 1$, i.e. $E_{rp} \approx E_{up}$. Since z is very small, $\dot{E} \approx \dot{z}$ and

$$\dot{z} \approx -Cz \quad \Rightarrow \quad z(t) = z(0)\,e^{-ct}.$$

Since $z(t)$ is an exponentially decaying function, it can never reach zero in finite time. Hence the error can never be zero in finite time. In addition, the control law is so aggressive that it may drive the cart out of the track while pumping the energy.

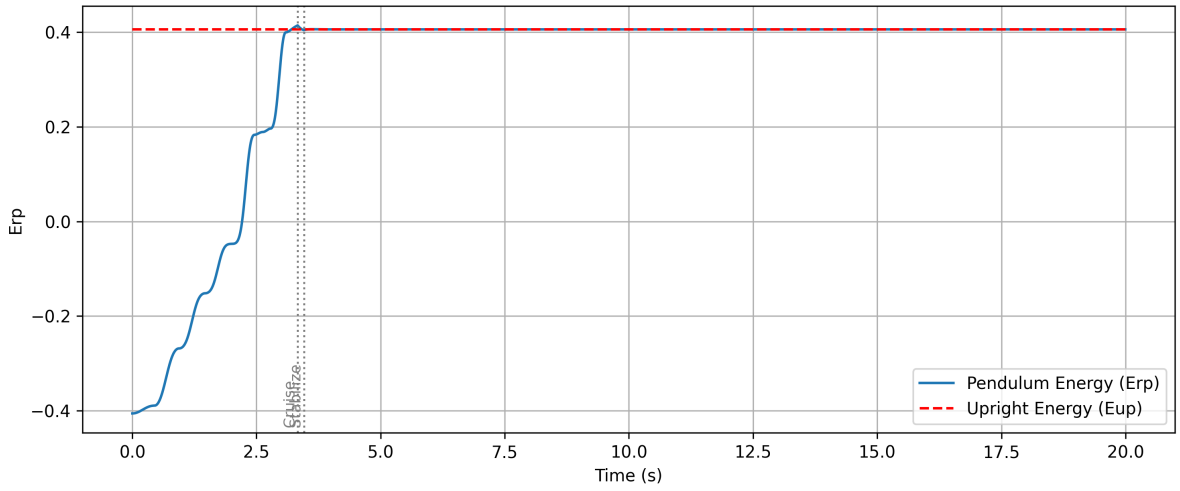


Figure 2: Energy evolution using the law proposed by Chatterjee *et al.* [2].

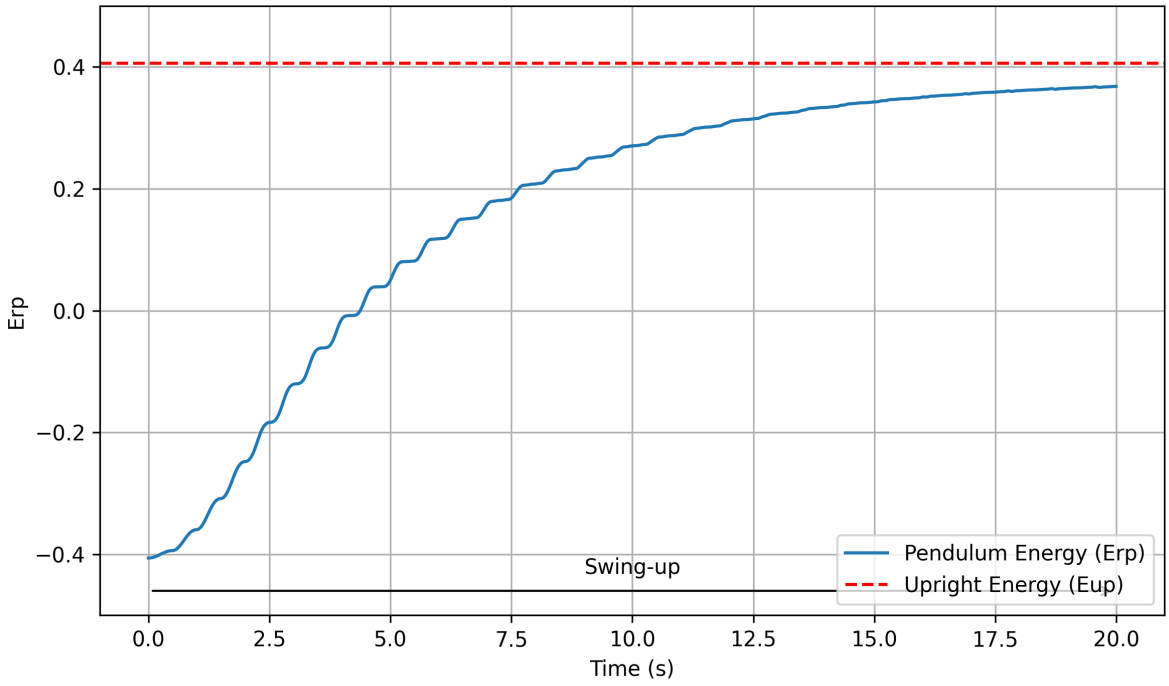


Figure 3: Energy evolution using the law proposed by Åström and Furuta [1].

4. Proposed Control Laws

For restricted track length, we introduce the concept of potential wells [2]. It is formulated in such a way that the cart experiences a repulsive force as it approaches the boundaries, in the neighbourhood of the limits:

$$\ddot{x}_{\text{cart-well}} = K_{cw}\ \text{sgn}(x)\ \log\left(1 - \frac{|x|}{L}\right).$$

The final equation after the formulation of the "cart potential well" is

$$\ddot{x} = -K_{su}\ \mathrm{sgn}(\dot{\theta}\cos\theta) + K_{cw}\ \mathrm{sgn}(x)\ \log\left(1 - \frac{|x|}{L}\right), \qquad K_{su},\ K_{cw} > 0.$$

A similar control law is developed for the cart velocity:

$$\ddot{x}_{\text{velocity-well}} = K_{vw}\ \mathrm{sgn}(\dot{x})\ \log\left(1 - \frac{|\dot{x}|}{\dot{x}_{\max}}\right),$$

where $\dot{x}_{\max}$ depends on the maximum velocity that the system can produce, or the velocity that the controller can handle.

A control law for energy maintenance can be formulated as

$$\ddot{x}_{\text{energy-maintain}} = K_{em}\left(e^{\,|E_{rp}-E_{up}|} - 1\right)\mathrm{sgn}(E_{rp} - E_{up})\ \mathrm{sgn}(\dot{\theta}\cos\theta).$$

The total acceleration is given as

$$\ddot{x}_{\text{Total}} = \ddot{x}_{\text{cart-well}} + \ddot{x}_{\text{velocity-well}} + \ddot{x}_{\text{energy-maintain}}.$$

For the energy-maintain law to function, E_{rp} must be different from E_{up}: as E_{rp} becomes equal to E_{up}, its contribution vanishes, so the energy will not be able to remain close to E_{up}. Therefore, a new law is formulated as

$$\ddot{x}_{em} = K_{em}\left(e^{\,|E_{rp}-\eta E_{up}|} - 1\right)\mathrm{sgn}(E_{rp} - E_{up})\ \mathrm{sgn}(\dot{\theta}\cos\theta), \qquad \eta > 1.$$

This ensures that the system remains close to the upright position until the controller catches the pendulum and stabilizes it at the upright position.

5. Stabilization

5.1 LQR Control

The LQR control technique is used for stabilization after linearizing the state-space equation about the unstable equilibrium $[0\ 0\ 0\ 0]^T$.

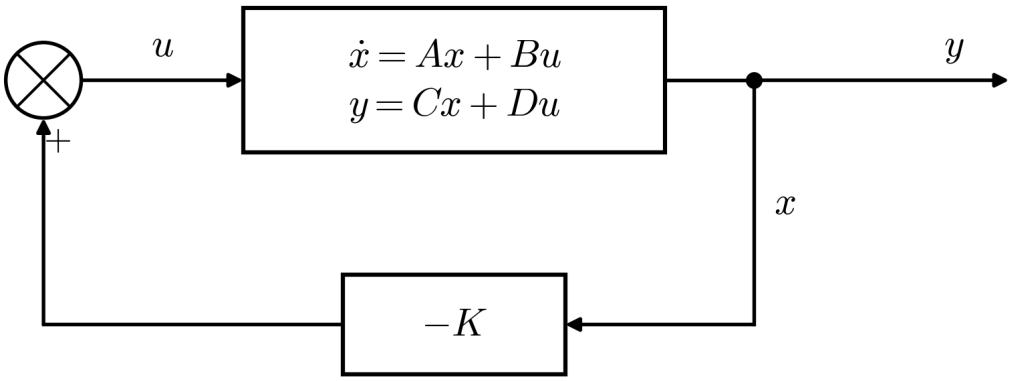


Figure 4: Block diagram of the LQR state-feedback loop.

The state-space model with control input u is

$$\dot{x} = Ax + Bu, \qquad x = [x_1,\, x_2,\, x_3,\, x_4]^T.$$

After the addition of state-feedback control $u = -Kx$, the state-space equation becomes

$$\begin{aligned} \dot{x} &= Ax + B(-Kx), \\ \dot{x} &= (A - BK)\, x, \end{aligned}$$

where

$$K = R^{-1}B^T P,$$

and P is the solution of the algebraic Riccati equation

$$PA + A^T P + Q - PBR^{-1}B^T P = 0,$$

with

$$Q = \begin{bmatrix} 100 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 50 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad R = 5.$$

We obtain

$$K = \begin{bmatrix} -4.4721,\ -75.9573,\ -7.1427,\ -11.5948 \end{bmatrix}.$$

5.2 Feedback Linearization

The non-linear model is written in the control-affine form below [3, 4]:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} x_3 \\ x_4 \\ \dfrac{-\dfrac{3mg \sin x_2 \cos x_2}{4(M_c + M_R)} + \dfrac{M_R l x_4^2 \sin x_2}{M_c + M_R}}{D(x_2)} \\ \dfrac{\dfrac{3g \sin x_2}{4l} - \dfrac{3M_R x_4^2 \sin x_2 \cos x_2}{4(M_c + M_R)}}{D(x_2)} \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \dfrac{1}{(M_c + M_R)\, D(x_2)} \\ \dfrac{-3 \cos x_2}{4l\, (M_c + M_R)\, D(x_2)} \end{bmatrix} F$$

where

$$D(x_2) = 1 - \frac{3M_R \cos^2 x_2}{4(M_c + M_R)}.$$

Consider this as a dynamical system of the form

$$\dot{x} = f(x) + g(x)\, u.$$

Let

$$y = h(x) = x_1, \qquad \dot{y} = \dot{x}_1 = x_3, \qquad \ddot{y} = \dot{x}_3.$$

The Lie derivatives are

$$\begin{aligned} L_f h &= [1\ 0\ 0\ 0]\, f(x) = x_3, \\ L_f(L_f h) &= [0\ 0\ 1\ 0]\, f(x) = \frac{-\dfrac{3mg \sin x_2 \cos x_2}{4(M_c + M_R)} + \dfrac{M_R l x_4^2 \sin x_2}{M_c + M_R}}{D(x_2)}, \\ L_g h &= [1\ 0\ 0\ 0]\, g(x) = 0, \\ L_g L_f h &= [0\ 0\ 1\ 0]\, g(x) = \frac{1}{(M_c + M_R)\, D(x_2)}. \end{aligned}$$

Then

$$\begin{aligned} y &= h(x), \\ \dot{y} &= L_f h + \underbrace{L_g h}_{0}\, u, \\ \ddot{y} &= L_f^2 h + L_g L_f h\, u. \end{aligned}$$

The relative degree is $r = 2$; hence the system is partially feedback linearizable. Define

$$z \triangleq \phi(x) = \begin{bmatrix} h(x) \\ L_f h(x) \\ \phi_1(x) \\ \phi_2(x) \end{bmatrix} = \begin{bmatrix} z_1 \\ z_2 \\ z_3 \\ z_4 \end{bmatrix}, \qquad \dot{z}_r = L_f^r h + L_g L_f^{r-1} h\, u.$$

Therefore,

$$\begin{bmatrix} \dot{z}_1 \\ \dot{z}_2 \end{bmatrix} = \begin{bmatrix} z_2 \\ L_f^2 h + L_g L_f h\, u \end{bmatrix}.$$

Setting

$$\dot{z}_2 = L_f^2 h + L_g L_f h\, u \triangleq V,$$

the control input is

$$u = F = \frac{1}{L_g L_f h}\Big(V - L_f^2 h\Big),$$

$$F = (M_c + M_R)\, D(x_2) \left(V + \frac{3mg \sin x_2 \cos x_2}{4(M_c + M_R)\, D(x_2)} - \frac{M_R l x_4^2 \sin x_2}{(M_c + M_R)\, D(x_2)} \right),$$

$$F = V\, D(x_2)\, (M_c + M_R) + \frac{3mg \sin x_2 \cos x_2}{4} - M_R l x_4^2 \sin x_2.$$

Normal-form coordinate transformation. The remaining coordinates must satisfy $L_g \phi_1 = L_g \phi_2 = 0$. Choose

$$\begin{aligned} z_1 &= h(x) = x_1, \\ z_2 &= \dot{x}_1 = x_3, \\ z_3 &= \phi_1(x) = x_2, \\ z_4 &= \phi_2(x) = \frac{3 \cos x_2}{4l}\, x_3 + x_4. \end{aligned}$$

The condition $L_g \phi_2 = 0$ gives

$$\begin{pmatrix} \dfrac{\partial \phi_2}{\partial x_1} & \dfrac{\partial \phi_2}{\partial x_2} & \dfrac{\partial \phi_2}{\partial x_3} & \dfrac{\partial \phi_2}{\partial x_4} \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ g_3 \\ g_4 \end{pmatrix} = 0,$$

$$\frac{\partial \phi_2}{\partial x_3} \left(\frac{1}{(M_c + M_R)\, D(x_2)} \right) + \frac{\partial \phi_2}{\partial x_4} \left(\frac{-3 \cos x_2}{4l\, (M_c + M_R)\, D(x_2)} \right) = 0,$$

$$\frac{\partial \phi_2}{\partial x_3} - \frac{3 \cos x_2}{4l}\, \frac{\partial \phi_2}{\partial x_4} = 0.$$

Transformed dynamics.

$$\begin{aligned}
\dot{x}_1 &= x_3, & \dot{x}_3 &= V,\\
\dot{z}_1 &= z_2, & \dot{z}_2 &= V,\\
z_3 &= x_2, & \dot{z}_3 &= \dot{x}_2 = x_4,\\
z_4 &= x_4 + \frac{3\cos x_2}{4l}\, x_3, & x_4 &= z_4 - \frac{3\cos x_2}{4l}\, x_3.
\end{aligned}$$

With $z_3 = x_2$ and $\dot{z}_3 = \dot{x}_2 = x_4$,

$$\begin{aligned}
\dot{z}_3 &= z_4 - \frac{3\cos z_3}{4l}\, z_2,\\
\dot{z}_4 &= \dot{x}_4 + \frac{3}{4l}\Big(-\sin x_2\, \dot{x}_2\, x_3 + \cos x_2\, \dot{x}_3\Big),\\
\dot{z}_4 &= \dot{x}_4 + \frac{3}{4l}\Big(-\sin x_2\, x_4\, x_3 + \cos x_2\, V\Big),\\
\dot{z}_4 &= \frac{3g\sin z_3}{4l\, D(x_2)} - \frac{3M_R x_4^2 \sin x_2 \cos x_2}{4(M_c + M_R)\, D(x_2)} - \frac{3}{4l}\sin x_2\, x_4\, x_3 + \frac{\cos x_2\, V}{4l},\\
\dot{z}_4 &= \frac{3g\sin z_3}{4l} - \frac{3\sin z_3}{4l}\, z_2 \left(z_4 - \frac{3\cos z_3}{4l}\, z_2\right).
\end{aligned}$$

In the new coordinates, the linear (controllable) subsystem is

$$\begin{bmatrix} \dot{z}_1 \\ \dot{z}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} V,$$

and the internal dynamics are

$$\begin{bmatrix} \dot{z}_3 \\ \dot{z}_4 \end{bmatrix} = \begin{bmatrix} z_4 - \dfrac{3\cos z_3}{4l}\, z_2 \\ \dfrac{3g\sin z_3}{4l} - \dfrac{3\sin z_3}{4l}\, z_2 \left(z_4 - \dfrac{3\cos z_3}{4l}\, z_2\right) \end{bmatrix},$$

where

$$z_1 = x_1, \qquad z_2 = x_3, \qquad z_3 = x_2, \qquad z_4 = \frac{3}{4l}\cos x_2 + x_4.$$

6. Simulation Results

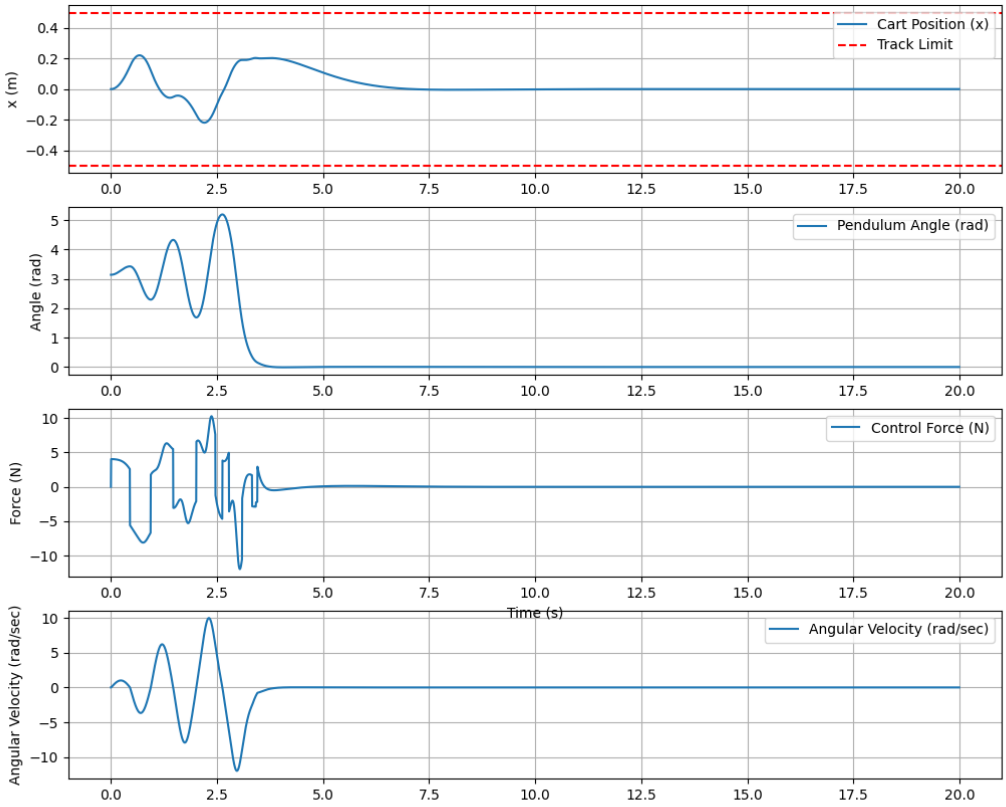


Figure 5: Simulation results: cart position with track limit, pendulum angle, angular velocity, and control force.

7. Stability Analysis

Two different Lyapunov functions are formulated, one for the swing-up mode and one for the cruise mode.

7.1 Swing-Up Mode

For the swing-up to be stable in the $[\theta, \dot{\theta}]^T$ subspace, there must exist a Lyapunov function $V_{\text{swing-up}}$ such that

$$V_{su} > 0 \ \ \forall\, x_{[\theta,\dot{\theta}]^T} \neq 0, \qquad V_{su}(0) = 0, \qquad \dot{V}_{su} \leq 0 \ \ \forall\, t > 0.$$

Choose

$$
\begin{aligned}
V_{su} &= E_{up} - E_{rp}, \\
\dot{V}_{su} &= -\dot{E}_{rp} = M_R \ddot{x}\, l\, \dot{\theta} \cos\theta, \\
\ddot{x} &= -K_{su} \operatorname{sgn}(\dot{\theta}\cos\theta) + K_{cw} \operatorname{sgn}(x) \log\left(1 - \frac{|x|}{L}\right), \\
\ddot{x} &= -K_{su} \operatorname{sgn}(\dot{\theta}\cos\theta) - \frac{d\phi}{dx}.
\end{aligned}
$$

Now,

$$
\begin{aligned}
\dot{V}_{su} &= M_R l\, \dot{\theta}\cos\theta \left(-K_{su} \operatorname{sgn}(\dot{\theta}\cos\theta) - \frac{d\phi}{dx}\right) \leq 0, \\
\dot{V}_{su} &= M_R l \Big(\operatorname{sgn}(\dot{\theta}\cos\theta)\, |\dot{\theta}\cos\theta|\Big) \left(-K_{su} \operatorname{sgn}(\dot{\theta}\cos\theta) - \frac{d\phi}{dx}\right) \leq 0, \\
\dot{V}_{su} &= M_R l\, |\dot{\theta}\cos\theta| \left(-K_{su} - \operatorname{sgn}(\dot{\theta}\cos\theta)\frac{d\phi}{dx}\right) \leq 0.
\end{aligned}
$$

Here, $M_R l > 0$, and $|\dot{\theta}\cos\theta|$ is positive and has no direction switching. Therefore,

$$
\begin{aligned}
-K_{su} - \operatorname{sgn}(\dot{\theta}\cos\theta)\frac{d\phi}{dx} &\leq 0, \\
K_{su} &\geq -\operatorname{sgn}(\dot{\theta}\cos\theta)\frac{d\phi}{dx}, \\
K_{su} &\geq -\left(\frac{d\phi}{dx}\right)\operatorname{sgn}(\dot{\theta}\cos\theta).
\end{aligned}
$$

This is the necessary condition for the Lyapunov function to be stable, because if

$$
K_{su} < \left(-\frac{d\phi}{dx}\right)\operatorname{sgn}(\dot{\theta}\cos\theta),
$$

then $\dot{V}_{su} > 0$. This means that the energy decreases and the swing-up fails.
From the Lyapunov function,

$$
\dot{V}_{su} = M_R l\, |\dot{\theta}\cos\theta| \left(-K_{su} - \operatorname{sgn}(\dot{\theta}\cos\theta)\frac{d\phi}{dx}\right) \leq 0.
$$

For this to be true for both swing directions,

$$
K_{su} \geq \left|\frac{d\phi}{dx}\right|, \qquad \text{i.e.,} \quad \text{(swing-up force)} \geq \text{(repulsive force)}.
$$

Intuitively, this condition must hold because if it is violated, the energy decreases and the swing-up fails.

7.2 Cruise Mode

For the cruise mode to be stable in the $[\theta, \dot{\theta}]^T$ subspace, there must exist a Lyapunov function V_{cruise} such that

$$V_{\text{cruise}} > 0 \;\; \forall\, x_{[\theta,\dot{\theta}]^T} \neq 0, \qquad V_{\text{cruise}}(0) = 0, \qquad \dot{V}_{\text{cruise}} \leq 0 \;\; \forall\, t > 0.$$

The Lyapunov function for this mode can be formulated as

$$\begin{aligned}
V_{\text{cruise}} &= \tfrac{1}{2}\Big(E_{rp} - E_{up}\Big)^2, \\
\dot{V}_{\text{cruise}} &= \left(E_{rp} - E_{up}\right) \dot{E}_{rp} = \left(E_{rp} - E_{up}\right)\Big(-M_R \ddot{x}\, l\, \dot{\theta} \cos\theta\Big), \\
\dot{V}_{\text{cruise}} &= \Big(-M_R \ddot{x}\, l\, \dot{\theta} \cos\theta\Big)\Big(E_{rp} - E_{up}\Big).
\end{aligned}$$

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